AI Content Performance Analysis Using Python and Power BI

1. Introduction

Artificial Intelligence (AI) is widely used for creating digital content such as text, images, and videos. This project analyzes the performance of AI-generated content using Python and Power BI. The aim is to understand content performance, compare platforms, and provide business insights using data analysis and visualization.

2. Objectives

- To analyze AI content performance.
- To clean and preprocess the dataset using Python.
- To handle missing values.
- To perform Exploratory Data Analysis (EDA).
- To build an interactive Power BI dashboard.
- To generate business insights and recommendations.

3. Dataset Description

The dataset contains information about AI-generated content published on different platforms.

Important columns include:

- Content_ID.
- Platform
- AI_Tool_Used
- AI_Type
- Content_Type
- Duration_Seconds
- Total_Views
- Total_Likes
- Total_Shares
- Total_Comments
- Engagement_Rate
- Production_Cost_USD
- Revenue_USD
- Watch_Time_Avg
- Retention_Rate

4. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Excel
- Power BI

5. Methodology

The project was completed in the following steps:

- Loaded the dataset into Python.
- Cleaned the data.
- Handled missing values.
- Checked data types.
- Performed Exploratory Data Analysis (EDA).
- Created Python visualizations.
- Exported the cleaned dataset to Excel.
- Imported the cleaned dataset into Power BI.
- Built an interactive dashboard.
- Generated business insights and recommendations.

6. Data Cleaning

The dataset was cleaned using Python.

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked missing values.
- Imputed missing values.
- Verified data types.
- Removed data inconsistencies.
- Prepared the dataset for visualization.

7. Python Visualizations

The following charts were created using Python:

understand the dataset before building the Power BI dashboard.

Brief Explanation

Count Plot – Displays the count of each category in the dataset.

Histogram Plot – Shows the distribution of numerical values.

Box Plot – Identifies outliers and shows data distribution.

Violin Plot – Displays both the distribution and density of data.

Bar Chart – Compares values across different categories.

Horizontal Bar Chart – Compares categories using horizontal bars.

Pie Chart – Shows the percentage share of different categories.

Scatter Plot – Shows the relationship between two numerical variables.

Correlation Heatmap – Displays the correlation between numerical variables.

Area Chart – Shows trends and cumulative values over data.

Pivot Table – Summarizes data for comparison and analysis.

Pivot Table Heatmap – Highlights important values in the pivot table using color intensity.

8. Power BI Dashboard

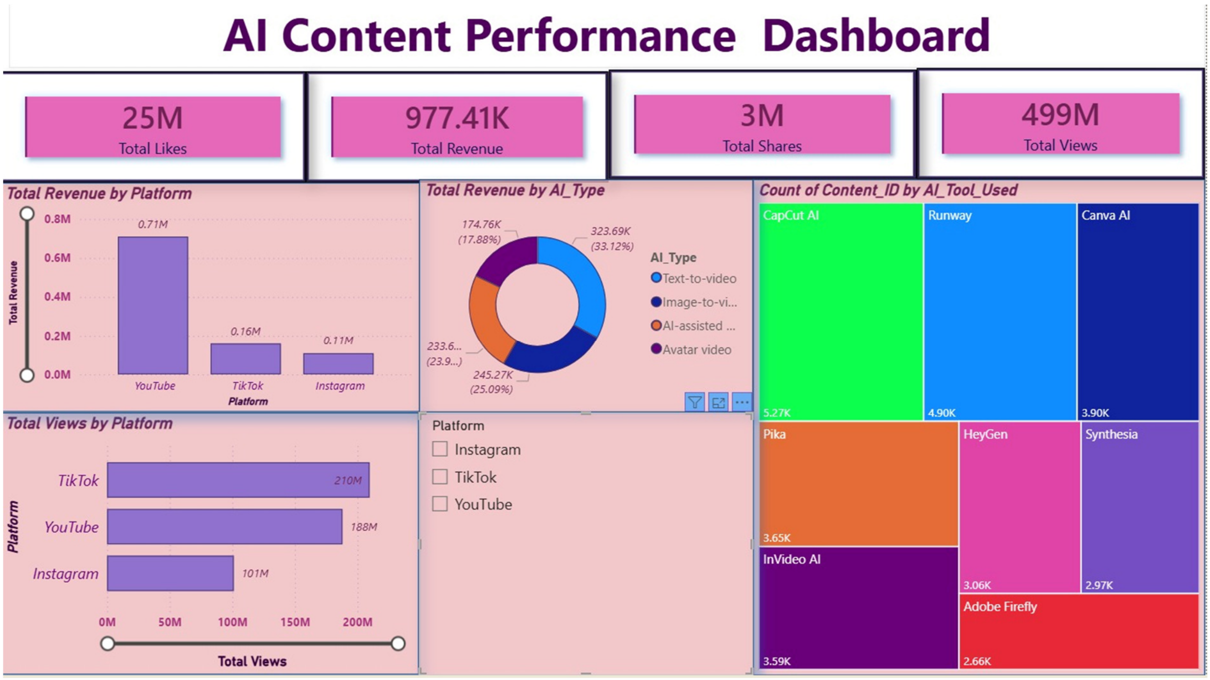
Two dashboards were created.

Dashboard 1 – AI Content Performance Dashboard

This dashboard includes:

- Total Views
- Total Likes
- Total Shares
- Total Revenue
- Total Views by Platform
- Total Revenue by Platform
- Revenue by AI Type
- Content Count by AI Tool

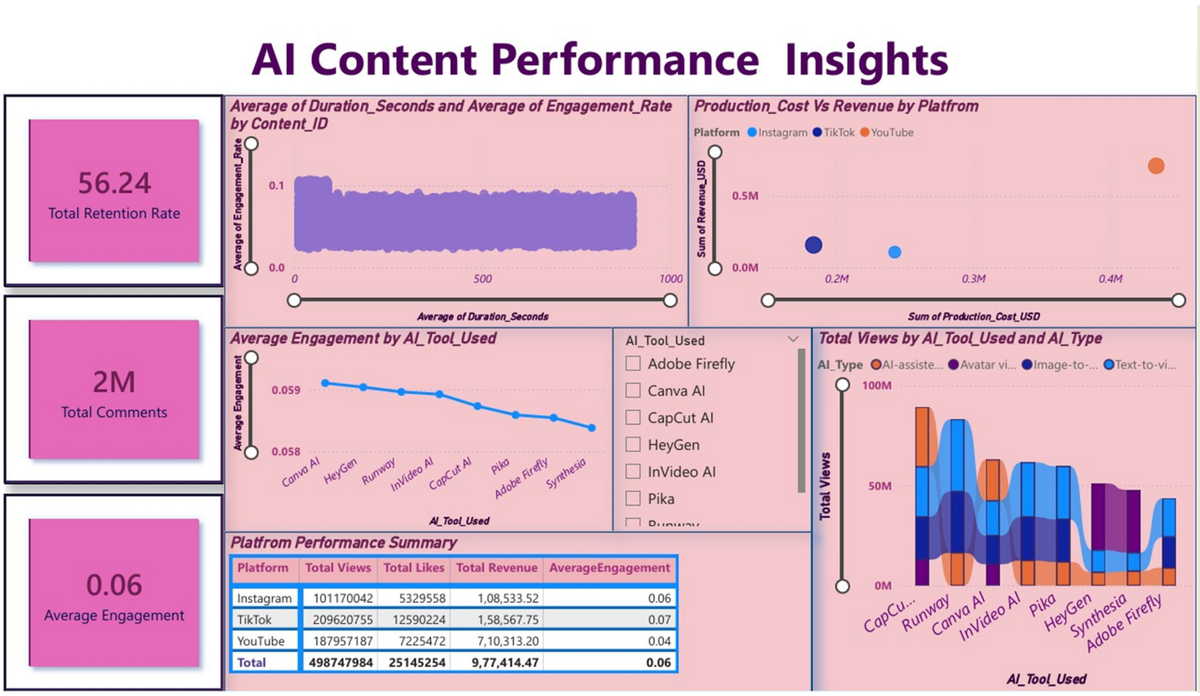
This dashboard provides an overview of AI content performance.



Dashboard 2 – AI Content Performance Insights

This dashboard includes:

- Duration vs Engagement Rate
- Production Cost vs Revenue
- Average Engagement by AI Tool
- Platform Performance Matrix
- This dashboard provides detailed business insights.



9. Business Insights

- Some platforms generated more views than others.
- Revenue varies across different platforms.
- AI tools have different engagement levels.
- Production cost influences revenue.
- Content duration affects engagement.
- Interactive dashboards help identify high-performing content.

10. Recommendations

- Focus on high-performing platforms.
- Use AI tools that provide better engagement.
- Optimize content duration.
- Improve production cost efficiency.
- Monitor dashboard metrics regularly.

11. Conclusion

This project successfully analyzed AI-generated content using Python and Power BI. Data cleaning, exploratory data analysis, and dashboard creation helped identify important business insights. The dashboard supports better decision-making by comparing platforms, AI tools, engagement, and revenue.